FIN2020 Midterm Exam

1. (25 points) True, False or it depends? If your answer is false or it depends, briefly explain your answer.

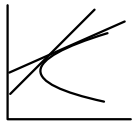
(a) Expected utility is invariant under any monotonic increasing transformation.

False. only under linear satisfy.

(b) If stock A First-Order Stochastic Dominance stock B', then asset A's expected return must be higher than asset B' expected return.

True

(c) Stock A has a higher Sharpe Ratio than stock B. Then all risk averse investors who prefer more to less would prefer stock A to B.



False. unknown $U_A < or > U_B$

(d) Asset 1's random return is r1 and asset 2's random return is r2. If $r1=r2+\theta$, and $E[\theta] = 0$, then asset 2 Second-Order Stochastic Dominance asset 1.

~~True~~ False. $r2 \perp \theta$

(e) If investor A is more risk averse than investor B, then for the asset with random return Z, investor A's certainty equivalent CEA(Z) must be higher than investor B's certainty equivalent CEB(Z).

Yes

2.(25 points) Consider an investor whose initial wealth is $10. Her Bernoulli utility is u(x)=x, for x≤10, 0.5x+5, for x≥10

This investor is offered two lotteries L1 = (-$3, $3, 0.5) and L2 = (-$6, $6, 0.5).

First judge whether the following statements are right or wrong. If the statement is right, please provide a brief explanation. If the statement is wrong, please give the correct statement.

(a) For $x \le 10$, $u'(x)=1$, $u''(x)=0$, $R_A(x)=0$
For $x \ge 10$, $u'(x)=0.5$, $u''(x)=0$, $R_A(x)=0$
True

$L1: \frac{1}{2}\times 7+(0.5\times 13+5)\times\frac{1}{2}=\frac{37}{4}$
$L2: \frac{1}{2}\times 4+(0.5\times 16+5)\times\frac{1}{2}=\frac{17}{2}$
$\therefore u(10)=10>\frac{1}{2}[u(L_1)+u(L_2)]=17.75$

(a) This investor is risk neutral.
False.

(b) If L1's reward is $0.7, then the investor is willing to participate.
(b) $E(x)=3\times 0.5-3\times 0.5=0<0.7$, the investor is willing to participate.

(c) In order to persuade the investor to participate in L2, L2's reward must be higher than L1's reward.
Yes, because the expected return is the same but L2 has higher variance. so the reward must be higher. risk averse

(d) If the investor's initial wealth is increased to 15, then this investor is indifferent between L1 and L2.
$L1: u(x)=(0.5\times 12+5)\times 0.5+0.5\times 14=12.5$
$L2: u(x)=0.5\times 9+0.5\times 15.5=12.25<12.5$
not indifferent.

(e) If the investor's initial wealth is increased to 20, then this investor is indifferent between L1 and L2.
$L1: \frac{1}{2}\times(0.5\times 17+5)+\frac{1}{2}\times(0.5\times 23+5)=15$
$L2: \frac{1}{2}\times(0.5\times 14+5)+\frac{1}{2}\times(0.5\times 26+5)=15$
$\therefore$ investor is indifferent between L1 and L2

3. (25 points) Suppose that a risk-averse investor with vNM expected utility and initial income Y_0 = 9 is offered a risky asset with random payoff Z that equals 7 in a good state that occurs with probability 1/2 and 0 in a bad state that occurs with probability 1/2.

(a)If this investor is offered a choice between the risky asset described above or a safe asset that pays off 3.5 no matter what, which one will he or she take: the risky asset or the safe asset? Because expected return is the same but risky asset has higher variance, he or she will choose safe asset.

(b)Recall that the certainty equivalent for CE(Z) is defined as the amount CE(Z) that makes the investor indifferent between taking the risky as. set and taking a safe asset that pays off CE(Z) for sure. Will CE(Z) be greater than, less than, or equal to 3.5? CE(Z) ~~greater~~ smaller than 3.5

(c) Now suppose that the investor has Bernoulli utility function of the form U(Y)= $Y^{0.5}$, where Y is his or her income in any given state. Use the information given above to calculate the numerical value of the certainty equivalent CE(Z) for the risky asset Z.

safety asset: $u(13.5) = 13.5^{0.5}$

risky asset: $\frac{1}{2}u(10+Z) + \frac{1}{2}u(10+Z) = \frac{1}{2}\left[(17+Z)^{0.5} + (10+Z)^{0.5}\right]$

$(9+CE)^{0.5} = 0.5\times(9+7)^{0.5} + 0.5\times(9+0)^{0.5}$

$\therefore CE = 3.25$

4. (25 points) Consider a consumer with initial wealth of 100. who faces a 1/5 (20 percent) chance of incurring a loss of 75 (which then brings his or her income down to 25). Suppose the consumers preferences are described by a VNM expected utility function with Bernoulli utility function $u(Y)=\frac{Y^{\gamma}-1}{1-\gamma}$, where, as we saw in class, $\gamma > 0$ measures the consumers constant coefficient of relative risk aversion.

(a)Assume that the investors coefficient of relative risk aversion is $\gamma = 1/2$. Find the maximum amount x* that the consumer will pay for an insurance policy that protects him or her fully against the loss.

(b)Now assume that, in addition to the 1/5 (20 percent) chance of a loss of 75, there is also a 1/5 (20 percent) chance of an even bigger loss of 100 (leaving the consumer with no income in this very bad state). Find the maximum amount that the consumer with $\gamma = 1/2$ will be willing to pay for insurance again all losses now.

(c) Finally, go back to the case from part (a), where there is just a 1/5 (20%) chance of a loss of 75. Suppose that instead of γ=1/2, we set γ as 2, calculate in this case.

**********Good Luck!**********

FIN2020 Midterm Exam Solution

1. (25 points) (a) F. Linear.

(b)T. F_A(x) $\leq$F_B(x)

(c) F. U(x) unknown UA < or >UB?

(d) F. r2 $\perp\theta$

2. (25 points) (-3,3,0.5) (-2.3, 3.7, 0.5)

E(u(L1))=0.5*7.7+0.5*(13.7*0.5+5)=9.775<10

u[(x1+x2)/2]> 1/2u(x1)+1/2u(x2)

(a) F (Risk averse)

(c) True (Risk Averse)

(d) E(u(L1))=12.25 L1 successor L2

(e) L1 ~ L2 E(u(L1))=E(u(L2))=15

3.Z=7, p=1/2 or 0, p=1/2

(a)E(Z)=0.5*7+0.5*0=3.5 safe asset

(b)CE<E(Z)=3.5

(c)(9+CE)^0.5=0.5$(9+7)^{0.5}$+0.5$(9+0)^{0.5}$.

CE=3.25

4.u(Y) = $\frac{Y^{\gamma-1}}{1-\gamma}$

(a) [$(100-x)^{0.5}$-1]/0.5=0.2*($25^{0.5}$-1)/0.5+0.8*($100^{0.5}$-1)/0.5.

x=19

(b) x=51